

1 Hyperbolic functions

Definition 1: hyperbolic sine function and hyperbolic cosine function

The hyperbolic functions:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \text{Sh } x = \frac{e^x - e^{-x}}{2} \quad (1)$$

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \text{Ch } x = \frac{e^x + e^{-x}}{2} \quad (2)$$

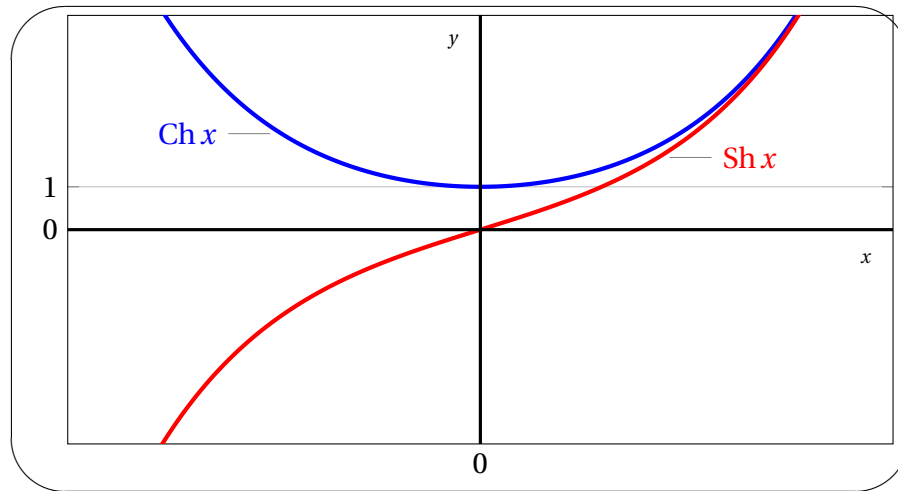
are called **hyperbolic sine** and **hyperbolic cosine**.

- The hyperbolic sine is an **odd** function, while the hyperbolic cosine is an **even** function. Hence we have:

$$\text{Sh}(-x) = -\text{Sh } x, \quad \forall x \in \mathbb{R} \quad \text{and} \quad \text{Ch}(-x) = \text{Ch } x, \quad \forall x \in \mathbb{R}$$

- We have the fundamental relations:

$$\text{Ch}^2 x - \text{Sh}^2 x = 1, \quad \forall x \in \mathbb{R} \quad \text{and} \quad \text{Sh } x \leq \frac{e^x}{2} \leq \text{Ch } x, \quad \forall x \in \mathbb{R} \quad (3)$$



- For the sign and monotonicity of the hyperbolic sine function we have:

$$\text{Sh } x = 0 \iff x = 0 \quad \begin{cases} \text{Sh } x > 0 & \text{if } x > 0 \\ \text{Sh } x < 0 & \text{if } x < 0 \end{cases}$$

f is increasing $\forall x \in \mathbb{R}$

- For the sign and monotonicity of the hyperbolic cosine function we have:

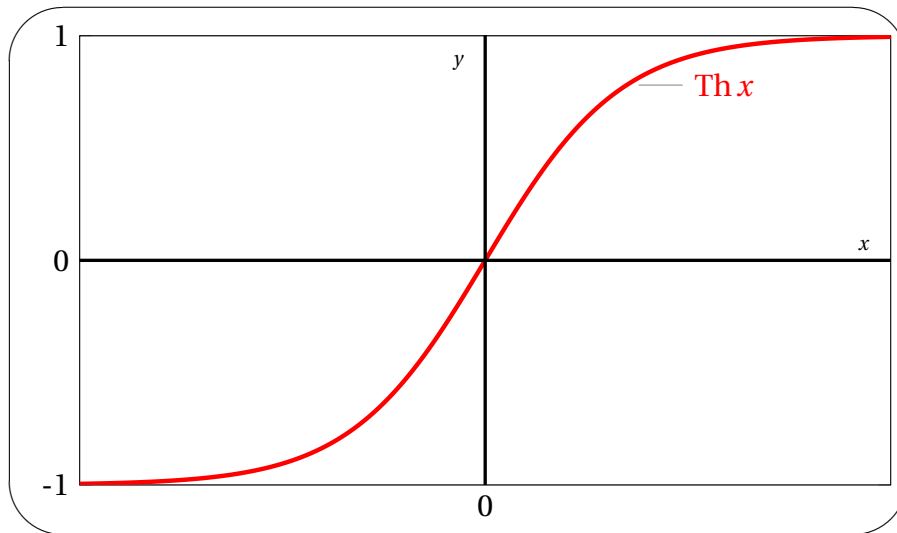
$$\text{Ch } x > 0, \quad \forall x \in \mathbb{R}$$

$$\begin{cases} \text{if } x < 0 & f \text{ is decreasing} \\ \text{if } x > 0 & f \text{ is increasing} \end{cases}$$

The hyperbolic function:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto \text{Th } x = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (4)$$

is called the **hyperbolic tangent** and it is an odd function.



2 Main hyperbolic formulas

Addition

$$\text{Sh}(x_1 + x_2) = \text{Sh } x_1 \text{ Ch } x_2 + \text{Sh } x_2 \text{ Ch } x_1 \quad (5a)$$

$$\text{Ch}(x_1 + x_2) = \text{Ch } x_1 \text{ Ch } x_2 + \text{Sh } x_1 \text{ Sh } x_2 \quad (5b)$$

$$\text{Th}(x_1 + x_2) = \frac{\text{Th } x_1 + \text{Th } x_2}{1 + \text{Th } x_1 \text{ Th } x_2} \quad (5c)$$

Double-argument formulas

$$\operatorname{Sh}(2x) = 2 \operatorname{Sh} x \operatorname{Ch} x \quad (6a)$$

$$\operatorname{Ch}(2x) = \operatorname{Ch}^2 x + \operatorname{Sh}^2 x \quad (6b)$$